1.3 Solving Two-Step Equations Using Properties of Equality

Lesson Introduction

Benchmark	MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical context or real-world context, where all terms are rational numbers.	Learning Targets	• I can relate properties of equality to solving two-step equations. • I can solve two-step equations in one variable.

	Building On	Working Toward
Benchmark Coherence	MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers. MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.	MA.8.AR.2.1 Solve multistep linear equations in one variable, with rational number coefficients. Include equations with variables on both sides.

Essential Questions	• How can I apply the properties of equality to solve two-step equations?
Lesson Narrative	In this lesson, students explore the properties of equality to build students' understanding of using the properties to solve two-step equations. Students notice that you can add, subtract, multiply, or divide both sides of an equation by the same number and the equation will still hold true. Then they apply that information to solving equations.

Component Summary	1.3.1 Warm-Up (~5 min.)	Notice and wonder about pictures of a seesaw *(SE p. 18)*	1.3.4 Guided Instruction (15 min.)	Solve two-step equations using properties of equality *(SE p. 21)*
	1.3.2 Exploration (15 min.)	Explore the properties of equality *(SE p. 19)*	1.3.5 Try It (10 min.)	Practice solving two-step equations using properties of equality *(SE p. 23)*
	1.3.3 Bring It Together (5 min.)	Review properties of equality *(SE p. 21)*	1.3.6 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 24)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: • Multiplication property of equality • Division property of equality • Subtraction property of equality • Addition property of equality • Symmetric property of equality	• Calculators • (Optional) Seesaw Video	• Consider creating and displaying an anchor chart with the properties of equality for students to reference. • The Warm-Up video is optional.

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may struggle to realize that multiplying by a reciprocal is more efficient.
Students may have trouble understanding why division by zero is undefined.
Students may not realize that the variable, $-x$, is multiplied by negative one, $-x = -1x$.</td><td>Consider providing students with sentence stems or answer choices for constructive response problems.
Allowing students to use calculators during the lesson will help students focus on discovery and process of using properties of equality to solve equations. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers.
Refrain from using the word “cancel.” Instead, say that the inverse operations add/subtract to 0 or multiply/divide to 1.
If time is an issue, then consider assigning 1.3.5 Try It for homework.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Notice and Wonder
In 1.3.2 Exploration, students are asked to complete two tables. In one table, they add, subtract, multiply, and divide the same amount to both sides of the equation. In the other table, they perform the same four basic operations but use different amounts on each side of the equation. Ask students what they notice and wonder about the tables.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
In 1.3.2 Exploration, students discuss and reflect on the properties of equality. Also, in 1.3.4 Guided Instruction, students discuss and reflect on each step they took to solve a two-step equation.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">In 1.3.5 Try It, consider replacing fractions and decimals with whole numbers so that students can focus on the process of solving a two-step equation rather than operations with rational numbers.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

1.3.1 Warm-Up: Notice and Wonder

Component Overview: *Students share what they notice and wonder about a series of pictures of a seesaw experiment. Ask students how the seesaw experiment relates to inequalities and equations.*

1.3.1 Warm-Up: Notice and Wonder

A math teacher and her daughter experimented with a seesaw at her local park. Their goal was to determine how the daughter could play on the seesaw on her own. She placed a crate on the other side of the seesaw and began adding bricks to it one at a time. Each time she added a brick, she tried to seesaw with the crate. A series of screenshots of her experiment are shown.

Seesaw with 0 Bricks	Seesaw after adding 1 Brick and a Crate
	
Seesaw after adding 12 Bricks and a Crate	
	

1. Complete the table to indicate one thing you notice and one thing you wonder when looking at this situation.

One thing I notice is...	One thing I wonder is...
Answers will vary. I notice that one brick is not heavy enough to lift the seesaw with the girl on it.	*Answers will vary. I wonder what it would look like with 6 bricks?*

2. Explain what is happening in this situation using the words "equal," "greater than," and/or "less than."
Answers will vary. In both pictures the girl's weight is never equal to the weight of the bricks. In the first picture with only one brick, the weight of the girl is greater than the weight of the brick. In the second picture with 12 bricks, the weight of the girl is less than the weight of the bricks.

The Warm-Up sets the stage for the students to think about what equality looks like. Solving equations using properties of equality means that the balance of the two expressions is kept intact. The visual idea of this balance can be referred to throughout the lesson. The 1.3.2 Exploration explores this mathematical balance and the imbalance of not using the properties. Questions that may be useful during the lesson:

- *What does the fulcrum of the seesaw represent?*
- *How does the balance in a seesaw relate to the balance of equations?*

1.3.2 Exploration: Properties of Equality

Component Overview: *Students explore the properties of equality. Students compare and contrast what happens when the same amount is added, subtracted, multiplied, or divided from both sides of an equation to what happens when different amounts are added, subtracted, multiplied, or divided from both sides of an equation.*

1.3.2 Exploration: Properties of Equality

1. The equation $-6+10=4$ is used in the table to show different operations.
   a. Work with your partner to complete the table.

Operation	Step	Show Step	Resulting Equation
$+$	Add 5 to both sides of the equal sign	$-6+10+5=4+5$	$9=9$
$-$	Subtract 8 from both sides of the equal sign	$-6+10-8=4-8$	$-4=-4$
$\times$	Multiply the terms on both sides of the equal sign by ½	$(-6+10)*\frac{1}{2}=4*\frac{1}{2}$	$2=2$
$\div$	Divide the terms on both sides of the equal sign by 2	$\frac{(-6+10)}{2}=\frac{4}{2}$	$2=2$

   b. What do you notice about each of the resulting equations?
   Both sides of the equation remain equivalent.
   c. With your partner, compare the results of multiplying each term in the original equation, $-6+10=4$, by $\frac{1}{2}$ and dividing each term by 2. Summarize your observations.
   When multiplying by $\frac{1}{2}$ and dividing by 2 you get the same answer.

Notice and Wonder
Ask students what they notice and wonder about the two tables. Students should notice that the equality between the left and right sides of the equation is preserved when they use the same value but not preserved when they use different values. Students may wonder if the properties hold for all rational numbers.

Highlight the equivalence of multiplying by a unit fraction and dividing by the denominator of the unit fraction (as seen in the multiplication row of the table).

2. Complete the statement.

   Dividing by 4 is the same as multiplying by $\frac{\boxed{1}}{\boxed{4}}$.

3. Use the equation $5\frac{1}{2} + 8 = 13.5$ to complete the following.

   a. Discuss with your partner why the equation could also be written $13.5 = 5\frac{1}{2} + 8$. Summarize your discussion.
   The equation could be written either way because as long as the terms remain equivalent it doesn't matter which side of the equal sign they are on.

   b. Work with your partner to complete the table for the equation $5\frac{1}{2} + 8 = 13.5$.

Operation	Step	Show Step	Resulting Equation
$+$	Add 23 to the left side of the equation and 14 to the right side	$5\frac{1}{2} + 8 + 23 = 13.5 + 14$	$36\frac{1}{2} = 27.5$
$-$	Subtract 5 from the left of the equation and 10 from the right side	$5\frac{1}{2} + 8 - 5 = 13.5 - 10$	$8\frac{1}{2} = 3.5$
$\times$	Multiply the terms on the left side of the equation by 3 and those on the right by -1	$3 * \left(5\frac{1}{2} + 8\right) = -1 * (13.5)$	$40\frac{1}{2} = -13.5$
$\div$	Divide the left side of the equal sign by 0 and the right side of the equal sign by 0.5	$\frac{5\frac{1}{2} + 8}{0} = \frac{13.5}{0.5}$	*undefined* $= 27$

Encourage students to practice reading the equation aloud from right to left and left to right.

Students may have been taught that you cannot divide by 0. The mathematically accurate statement is that division by 0 is undefined. If students are confused by this, then refer to how the properties of equality allowed them to define something that is true in mathematics. Mathematicians cannot define division by 0. The reason for no definition is best illustrated with an example. For example, in $0\overline{)56}$, ask students how many times 0 goes into 56. Any value they give can be used. The result is the dividend did not change. This helps to show students that 56 cannot be grouped into 0 groups.

1.3.2 Exploration: Properties of Equality (continued)

c. Prisha said it is not possible to complete the last row of the table. Explain why Prisha said it is not possible.
Prisha said that it is not possible to complete the last row of the table because division by 0 is undefined..

d. What do you notice about the resulting equations in the first three rows of the table?
The two sides of the equation are no longer equal.

4. Complete the statements.

An equation compares two equivalent expressions using the mathematical symbol = to show the values of both quantities are [○ equal. ○ unequal.] When the same operation is performed on both sides of an equation, the quantities [○ remain ○ are no longer] equivalent. When the operations performed on both sides of an equation are not the same, the resulting equation is [○ still ○ no longer] true because the quantities being compared are [○ still ○ no longer] equivalent.

Use talk moves to position students as sensemakers.

- *In your own words, how does performing operations impact the equivalence?*
- *Provide an example. What if we subtract 3 from one side and add 3 to the other side to keep it 'balanced'? Are they still equivalent? Why or why not?*

1.3.3 Bring It Together: Properties of Equality

Component Overview: *Students review the properties of equality and provide numeric examples of each property. Consider creating and displaying an anchor chart with the properties for students to reference throughout the remainder of the lesson.*

1.3.3 Bring It Together: Properties of Equality

Properties of equality can be used to solve equations.

1. Complete the table with a numeric example of each property of equality.

Property	Algebraic Statement	Numeric Example
Symmetric Property of Equality	If $a = b$, then $b = a$.	If $2 = 2$ Then $2 = 2$
Addition Property of Equality	If $a = b$, then $a + c = b + c$.	If $2 = 2$ Then $2 + 3 = 2 + 3$ $5 = 5$
Subtraction property of Equality	If $a = b$, then $a - c = b - c$.	If $2 = 2$ Then $2 - 3 = 2 - 3$ $-1 = -1$
Multiplication Property of Equality	If $a = b$, then $a \cdot c = b \cdot c$.	If $2 = 2$ Then $2 * 3 = 2 * 3$ $6 = 6$
Division Property of Equality	If $a = b$ and $c \neq 0$, then $a \div c = b \div c$.	If $2 = 2$ Then $\frac{2}{3} = \frac{2}{3}$

2. Explain why $c \neq 0$ for the division property of equality.
   c cannot equal 0 for the division property of equality because if you divide a number by zero it is undefined.

3. Which property justifies the equation $2m - 7.3 = 24.45$ is equivalent to $24.45 = 2m - 7.3$?
   The symmetric property of equality.

Students should be familiar with these properties both informally and formally. Consider asking students to provide rational numbers represented by decimals or fractions as examples. Division by 0 is emphasized again. Encourage students to note that division by 0 is undefined. Name the properties as they are used and ask students to name the properties as they use them. Once students show that they understand the properties intrinsically, then release them from having to state the properties.

1.3.4 Guided Instruction: Solving Two-Step Equations Using Properties of Equality

Component Overview: *Students practice solving two-step equations using properties of equality. Students are asked to identify which property they used.*

1.3.4 Guided Instruction: Solving Two-Step Equations Using Properties of Equality

When solving equations, the goal is to isolate the variable to find the value(s) of the variable that make the equation true.

The operations being performed on the variable can be "undone" using inverse operations and opposites in order to isolate the variable. Properties of equality can be applied to create equivalent equations using steps to isolate the variable.

1. Consider the equation $5x - 9.6 = 21.4$.

   a. Complete the statement to describe the equation. The variable, x, is first being ___*multiplied*___ by 5, and then 9.6 is being ____*subtracted*_____ from this product. The value of the expression, $5x - 9.6$, is ___*equal*___ to 21.4.

   b. Describe the operation that could be performed to isolate $5x$ on the left side of the equation.
   To isolate $5x$ you would add 9.6 to both sides of the equation.

   c. Show this step to create an equivalent equation.

   $$5x - 9.6 + 9.6 = 21.4 + 9.6$$
   $$5x = 31$$

   d. Describe the operation that could be performed to isolate the x on the left side of the equation.
   To isolate x you would divide both sides of the equation by 5

   e. Show this step to create an equivalent equation.

   $$5x \div 5 = 31 \div 5$$
   $$x = 6.2$$

Consider providing students with a word bank for question 1a. Including a symbol with each word will help ELL students associate the words with the mathematical symbols.

1.3.4 Guided Instruction: Solving Two-Step Equations Using Properties of Equality (continued)

f. What is the coefficient of the x after dividing both sides?
The coefficient of the x is 1.

g. Use substitution in the original equation to verify the solution.

$$5x - 9.6 = 21.4$$

$5(6.2) - 9.6 = 21.4$
$31 - 9.6 = 21.4$
$21.4 = 21.4$

h. Which properties of equality were applied to solve this equation?
The addition property of equality was used.
The division property of equality was also used.

i. Cameron said the equation $5x = 31$ could have been solved by applying the multiplication property of equality. What value can be used as the factor that is multiplied to both sides of the equation to isolate x?
$\frac{1}{5}$

2. Solve $-6.7 = \frac{y}{^-4} - \frac{1}{2}$ by applying properties of equality.

a. Fill in each box with an operation and each blank with a value.

$$-6.7 = \frac{y}{^-4} - \frac{1}{2}$$

$$-6.7 \boxed{+} \frac{1}{2} = \frac{y}{^-4} - \frac{1}{2} \boxed{+} \frac{1}{2}$$

$$-6.2 = \frac{y}{^-4}$$

$$-6.2 \boxed{\cdot} -4 = -4 \boxed{\cdot} \frac{y}{^-4}$$

$$24.8 = y$$

In general, the addition property of equality and the subtraction property of equality can be interchanged. For example, if students subtracted -9.6 from both sides, then the student could state that they used the subtraction property of equality. If a student multiplied both sides by $\frac{1}{5}$, then the multiplication property of equality would be stated. The Guided Instruction is meant to give students the understanding of the mathematics that justifies the steps in solving equations. The justification will depend upon the mathematics that were applied.

Refrain from using the word "cancel." Instead, say the property of equality that adds/subtracts to zero or multiplies/divides to one.

1.3.4 Guided Instruction: Solving Two-Step Equations Using Properties of Equality (continued)

b. Use substitution in the original equation to verify the solution.

$$-6.7 = \frac{y}{-4} - \frac{1}{2}$$

$$-6.7 = \frac{24.8}{-4} - \frac{1}{2}$$

$$-6.7 = -6.2 - \frac{1}{2}$$

$$-6.7 = -6.7$$

c. Which properties of equality were applied to solve this equation?
The addition property of equality was used.
The multiplication property of equality was also used.

Consider asking students to identify where each property was used in their work. They could underline, circle, or highlight where they used each property and then write the name of the property.

1.3.5 Try It: Solving Two-Step Equations Using Properties of Equality

Component Overview: *Students practice solving two-step equations using properties of equality. Remind students to check their solutions using substitution.*

1.3.5 Try It: Solving Two-Step Equations Using Properties of Equality

Work with your partner to complete the following.

1. Use the equation $4.76 = -5 + 3.2n$ to complete the following.

   a. Apply properties of equality to solve the equation.
   $4.76 + 5 = -5 + 5 + 3.2n$
   $9.76 = 3.2n$
   $\frac{9.76}{3.2} = \frac{3.2n}{3.2}$
   $3.05 = n$

   b. Use substitution to verify the solution.
   $4.76 = -5 + 3.2(3.05)$
   $4.76 = -5 + 9.76$
   $4.76 = 4.76$

2. Use the equation $\frac{5}{7}r + 6\frac{3}{7} = {}^{-}8$ to complete the following.

   a. Apply properties of equality to solve the equation.
   $\frac{5}{7}r + 6\frac{3}{7} - 6\frac{3}{7} = -8 - 6\frac{3}{7}$
   $\frac{5}{7}r = -14\frac{3}{7}$
   $\frac{7}{5} * \frac{5}{7}r = -14\frac{3}{7} * \frac{7}{5}$
   $r = -20\frac{1}{5}$

   b. Use substitution to verify the solution.
   $\frac{5}{7}\left(-20\frac{1}{5}\right) + 6\frac{3}{7} = -8$
   $-14\frac{3}{7} + 6\frac{3}{7} = -8$
   $-8 = -8$

Students could multiply by the reciprocal of $\frac{5}{7}$ in order to isolate the variable r. Consider engaging students with a discussion about what the reciprocal is and why multiplying by the reciprocal is more efficient than multiplying by seven and then dividing by five (two steps versus one step). Remind students of the existence of multiplicative inverses. ($a \times \frac{1}{a} = 1$ or $\frac{a}{b} \cdot \frac{b}{a} = 1$)

1.3.6 Wrap-Up: Cool Down

***Component Overview**: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

1.3.6 Wrap-Up: Cool Down

1. How can you use substitution to ensure your answer is correct?

   Answers will vary.

2. When solving an equation, why do operations have to be performed on both sides of the equal sign?

   Answers will vary.